Cloud Watch

1. **Definition**

CloudWatch is a monitoring and management service provided by Amazon Web Services that helps to monitor AWS resources and applications in real-time.

2. Key Features of CloudWatch

- **Real-time Monitoring: Tracks performance of AWS services.**
- **Metrics Collection: Collects data like CPU usage, memory, disk I/O.**
- **Log Management: Stores and analyzes log files.**
- **Alarms: Sends alerts when thresholds are crossed.**
- **Automatic Scaling Support: Works with auto scaling to adjust resources.**

3. Working of CloudWatch

1. **AWS resources send metrics data to CloudWatch.**
2. **CloudWatch stores and analyzes the data.**
3. **User sets thresholds (alarms).**
4. **When threshold is crossed, notification is triggered.**

4. Architecture (Text Diagram)

```
AWS Resources (EC2, S3, etc.)
          v
      CloudWatch
   (Metrics & Logs)
          v
    Alarms & Alerts
          v
     User / Admin
```

5. Advantages

- **Centralized monitoring of all AWS services**
- **Helps in performance optimization**
- **Improves system reliability**
- **Enables quick issue detection**

6. Example

In Amazon EC2, CloudWatch monitors CPU utilization. If CPU usage exceeds 80%, it sends an alert to the administrator.

Matrices

1. Definition

Metrics in Amazon CloudWatch are numerical data points that represent the performance and health of AWS resources over time.

2. Key Characteristics of Metrics

- **Time-based data (measured over intervals)**
- **Stored automatically by CloudWatch**
- **Used for monitoring and analysis**
- **Can be custom or default metrics**

3. Types of Metrics

1. **Default Metrics**
    - **Provided by AWS services automatically**
    - **Example: CPU Utilization, Network Traffic**
2. **Custom Metrics**
    - **Created by users for specific needs**
    - **Example: Number of active users in an application**

4. Common AWS Metrics Examples

- **In Amazon EC2: CPU usage, Disk read/write**
- **In Amazon S3: Number of requests, storage size**
- **In AWS Lambda: Function execution time**

5. Working of Metrics

1. **AWS resources generate performance data**
2. **Data is sent to CloudWatch as metrics**
3. **Metrics are stored and visualized using graphs**
4. **Used to create alarms and notifications**

6. Advantages

- **Helps in performance monitoring**
- **Supports decision making**
- **Enables automatic alert system**
- **Improves system efficiency**

7. Example

If CPU utilization of an EC2 instance increases continuously, metrics help identify the issue and take action like scaling resources.

Alarm & notification

1. Definition

Alarm in Amazon CloudWatch is a feature that monitors metrics and triggers actions when a defined threshold is crossed.

Notification is the alert message sent to users when the alarm is triggered.

2. Purpose

- **To detect system issues automatically**
- **To inform users about abnormal conditions**
- **To take immediate corrective actions**

3. Working of Alarm

1. **Select a metric (e.g., CPU usage)**
2. **Set a threshold value (e.g., 80%)**
3. **CloudWatch continuously monitors the metric**
4. **If threshold is crossed → Alarm state changes**
5. **Notification is sent to user**

4. Alarm States

- **OK – System is normal**
- **ALARM – Threshold exceeded**
- **INSUFFICIENT DATA – Not enough data available**

5. Notification System

- **Uses Amazon SNS (Simple Notification Service)**
- **Sends alerts via:**
    - **Email**
    - **SMS**
    - **Push notification**

6. Actions Triggered by Alarm

- **Send notification to admin**
- **Auto scaling (increase/decrease resources)**
- **Stop/terminate instances**

7. Example

In Amazon EC2, if CPU usage goes above 80%, CloudWatch triggers an alarm and sends an email notification to the administrator.

Log & billing Monitoring Other AWS monitoring

1. Log Monitoring

Definition : Log Monitoring in Amazon CloudWatch is used to collect, store, and analyze log data generated by AWS resources and applications.

Features

- **Stores logs using CloudWatch Logs**
- **Supports real-time log monitoring**
- **Provides search and filter options**
- **Helps in troubleshooting and debugging**
- **Can create log-based alarms**

Working

1. **AWS services generate logs**
2. **Logs are sent to CloudWatch**
3. **Logs are stored in log groups**
4. **User analyzes logs or sets alerts**

Example

In Amazon EC2, server logs are monitored to detect system errors or unauthorized access.

2. Billing Monitoring

Definition : Billing Monitoring is used to track AWS usage and control costs.

Features

- **Tracks monthly AWS charges**
- **Allows setting billing alarms**
- **Sends notifications when cost exceeds limit**
- **Helps in budget management**

Working

1. **AWS calculates usage cost**
2. **Data is sent to CloudWatch**
3. **User sets cost threshold**
4. **Notification is triggered if exceeded**

Example : If AWS usage cost exceeds ₹10,000, a notification is sent to the administrator.

3. Other AWS Monitoring Services

(a) AWS CloudTrail

- **Records API calls and user activities**
- **Used for security auditing and compliance**
- **Helps track who did what in AWS**

(b) AWS X-Ray

- **Monitors application performance**
- **Provides request tracing**
- **Helps identify bottlenecks and errors**

(c) AWS Trusted Advisor

- **Gives best practice recommendations**
- **Improves security, cost, and performance**
- **Suggests unused resource removal**

(d) Amazon Inspector

- **Performs security assessment of applications**
- **Detects vulnerabilities AND Improves system security**

Advantages

- **Helps in error detection and debugging**
- **Enables cost control and optimization**
- **Improves security and performance**

  Provides centralized monitoring system

Eucalyptus

1. Definition

Eucalyptus (Elastic Utility Computing Architecture for Linking Your Programs to Useful Systems) is an open-source cloud computing platform used to create private and hybrid clouds.

2. Key Features

- **Open-source and freely available**
- **Compatible with Amazon Web Services (AWS API)**
- **Supports private and hybrid cloud environments**
- **Provides scalability and flexibility**
- **Enables virtual machine management**

3. Architecture (Components)

1. **Cloud Controller (CLC)**
    - **Main controller that manages entire cloud**
2. **Cluster Controller (CC)**
    - **Manages a group of nodes**
3. **Node Controller (NC)**
    - **Controls virtual machines on physical servers**
4. **Walrus Storage**
    - **Provides storage similar to S3**
5. **Storage Controller (SC)**
    - **Manages block storage similar to EBS**

Text Diagram

```
Cloud Controller (CLC)
        |
Cluster Controller (CC)
        |
Node Controller (NC)
        |
 Virtual Machines
```

4. Working

1. **User sends request to Cloud Controller**
2. **Controller forwards request to Cluster Controller**
3. **Cluster Controller selects Node Controller**
4. **Node Controller runs Virtual Machines**
5. **Storage services provide required data**

5. Advantages

- **Cost-effective (open source)**
- **Easy to deploy private cloud**
- **AWS compatibility**
- **High control over data and security**

6. Limitations

- **Requires technical expertise**
- **Limited community support compared to commercial clouds**

7. Example : A university can use Eucalyptus to create a private cloud for students to run virtual machines for practical experiments without using public cloud services.

Microsoft Azure

1. Definition : Microsoft Azure is a commercial cloud computing platform provided by Microsoft that offers on-demand computing services like storage, networking, databases, and virtual machines.

2. Key Features

- **Provides Infrastructure as a Service (IaaS) and Platform as a Service (PaaS)**
- **Supports multiple programming languages (Java, .NET, Python)**
- **High scalability and availability**
- **Integrated with Microsoft tools like Windows Server**
- **Strong security and compliance**

3. Architecture / Components

1. **Compute Services**
    - **Virtual Machines, App Services**
2. **Storage Services**
    - **Blob Storage, Disk Storage**
3. **Networking**
    - **Virtual Network, Load Balancer**
4. **Database Services**
    - **SQL Database, Cosmos DB**
5. **Management Tools**
    - **Azure Portal, Monitoring tools**

Text Diagram

```
        User / Application
                |
          Azure Portal
                |
        --------------------------
   |   Compute    |  Storage   |
   |  Networking | Database   |
         --------------------------
                |
          Physical Data Centers
```

4. Working

1. **User accesses Azure through portal or API**
2. **Selects required services (VM, storage, etc.)**
3. **Azure allocates resources from data centers**
4. **Application runs on cloud infrastructure**
5. **User pays based on usage**

5. Advantages

- **Pay-as-you-use pricing**
- **High availability and reliability**
- **Easy integration with Microsoft products**
- **Global data centers**

6. Limitations

- **Cost can increase with high usage**
- **Requires internet connectivity**
- **Learning curve for beginners**

7. Example

A company hosts its web application on Azure Virtual Machines and stores data in Azure SQL Database to ensure scalability and global access.

Amazon EC2

1. Definition

Amazon EC2 (Elastic Compute Cloud) is a commercial cloud service provided by Amazon Web Services that offers resizable virtual servers (instances) for computing in the cloud.

2. Key Features

- **Provides virtual machines (instances)**
- **Scalable – increase or decrease resources anytime**
- **Pay-as-you-use pricing**
- **Supports multiple operating systems (Linux, Windows)**
- **Integration with other AWS services**

3. Architecture / Components

1. **EC2 Instance**
   - **Virtual server for running applications**
2. **AMI (Amazon Machine Image)**
   - **Pre-configured template for instances**
3. **Instance Types**
   - **Different CPU, memory configurations**
4. **EBS (Elastic Block Store)**
   - **Storage for instances**
5. **Security Groups**
   - **Acts as firewall for instances**

Text Diagram

```
    User Request
         |
     AWS Cloud
         |
    Amazon EC2
         |
  Virtual Instances
         |
   Storage (EBS)
```

4. Working

1. **User selects an AMI**
2. **Chooses instance type**
3. **Launches EC2 instance**
4. **Instance runs application**
5. **User can scale resources as needed**

5. Advantages

- **High scalability and flexibility**
- **Cost-effective (pay per use)**
- **Easy deployment of applications**
- **Reliable and secure**

6. Limitations

- **Cost management required**
- **Requires technical knowledge**
- **Dependent on internet connectivity**

7. Example

A startup hosts its website on EC2 instances. During high traffic, it increases instances to handle load and reduces them later to save cost.